Electric Forces: Guided Notes

Chemistry · Unit 1B · Level 3 (closer to the class work) · goes with L1.7-L1.9 (Coulomb's Law, Neutral Interactions, Static Interaction Review)

Name: _____

Use the class slides and your reading. Key words:

electric force | *attract* | *repel* | *Coulomb's law* | *distance* | *amount of charge* | *scientific notation* | *exponent* | *polarized* | *neutral*

A. Attract or repel (L1.7, L1.9)

1. Opposite charges _____. Like charges _____.

2. Two balloons are both rubbed on hair. Predict what happens when they hang near each other, and explain.

_____

_____

B. Coulomb's law: proportional reasoning (L1.7)

3. More charge makes the force _____. More distance makes the force _____.

4. **Charge rule: multiply the force by the same number as the charge.**

> Example: The force is 6 mN. One charge doubles (× 2). New force = 6 × 2 = 12 mN.

a) Force is 5 mN. One charge triples (× 3). New force = _____

b) Force is 8 mN. One charge is cut in half (÷ 2). New force = _____

5. **Distance rule: square the change, then divide.**

> Example: The force is 36 mN. The distance doubles (× 2). 2 × 2 = 4, so new force = 36 ÷ 4 = 9 mN.
> If the distance gets smaller, multiply instead: half the distance → force × 4.

a) Force is 20 mN. Distance doubles. New force = _____

b) Force is 18 mN. Distance triples. New force = _____

c) Force is 3 mN. Distance is cut in half. New force = _____

C. Scientific notation (L1.7)

6. **Write the number out: move the decimal to the right as many places as the exponent.**

> Example: $2 \times 10^3 = 2{,}000$ $4.5 \times 10^4 = 45{,}000$

a) $6 \times 10^5 =$ _____

b) $2.3 \times 10^3 =$ _____

c) Which is more charge: 3×10^{12} e or 8×10^9 e? _____

D. Neutral objects (L1.8)

7. A neutral object near a charged object becomes _____.

8. Draw a negative balloon next to a neutral piece of paper. Show where the paper's electrons move. Draw an arrow for the force.